
On the Uncertainty in Atomistic Diffusion Models

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Abstract

Diffusion models are state-of-the-art for generating atomistic structures, yet they still produce physically invalid samples. While uncertainty quantification (UQ) could mitigate this, traditional methods remain expensive due to the sequential nature of diffusion models. In this work, we probe the efficacy of a lightweight UQ-head, akin to heteroskedastic regression. We systematically leverage this UQ-head across three granularities: post-hoc filtering and two UQ-conditioned sampling interventions (molecule correction and atom forcing). We also show that chemical composition strongly confounds uncertainty, and therefore propose a normalised UQ-score to mitigate this bias. After empirically establishing that our scores reliably predict chemical validity, we evaluate how each mechanism, using both standard and normalised uncertainty, impacts sample quality, providing critical insights into the practical trade-offs of UQ-guided atomistic structure generation.

1 Introduction

Sampling-based generative models are now state-of-the-art across diverse fields, from image and audio synthesis to scientific domains like molecular generation [1, 2, 3, 4]. Despite their remarkable success, these models still occasionally produce samples of inconsistent or degraded quality, such as chemically invalid or unstable structures [5, 6]. To address this, recent work has started adapting uncertainty quantification techniques from traditional supervised learning to sampling-based generative models [7, 8, 9]. Just as predictive uncertainty helps identify unreliable outputs in discriminative tasks, generative uncertainty can help flag low-quality generations [10]. Beyond simple post-hoc detection, uncertainty can also be leveraged dynamically during the generative process to actively guide and improve sample quality [11]. Applying these techniques in practice, however, presents computational challenges. Because the generative process is inherently sequential, actively guiding the model requires an uncertainty estimate at every sampling step. This overhead can make traditional Bayesian or ensembling approaches computationally prohibitive. Therefore, there is a distinct need for lightweight, cheap-to-compute uncertainty estimates tailored for sequential generation.

In this work, we equip atomistic diffusion models with a heteroskedastic uncertainty head [12], and leverage it to (1) filter high-uncertainty samples post-hoc, (2) dynamically trigger Langevin correction steps during sampling, and (3) *force* confident atoms ahead in time to provide context for uncertain ones, also during sampling. Schusterbauer et al. [13] recently proposed the latter mechanism in an image setting. Fig. 1 provides an overview of the three mechanisms. Finally, we introduce a normalised UQ-score that aims to remove confounding effects stemming from molecule size and chemical composition.

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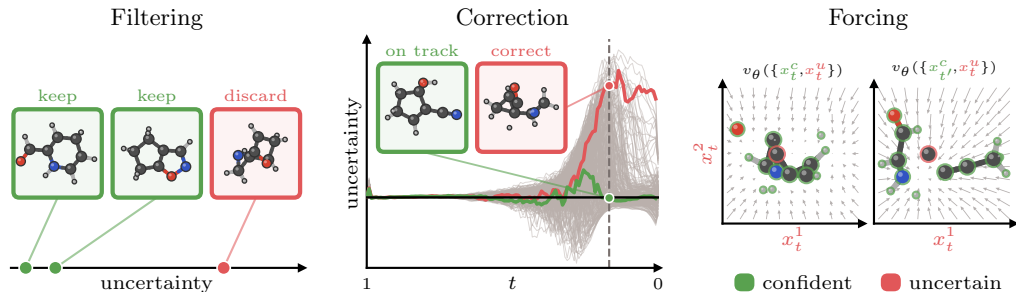


Figure 1: Uncertainty in atomistic diffusion models can play two roles, which we probe through three mechanisms: a post-hoc signal for filtering unreliable samples, and a guide used during generation to improve sample quality, through molecule-level trajectory correction or atom-level forcing.

Contributions Our contributions are summarised as follows:

- (i) We analyse how UQ can be leveraged at three different levels in atomistic diffusion models.
- (ii) We introduce a normalised UQ-score enabling efficient triggering of correction steps.
- (iii) We analyse the behaviour of the UQ-based samplers, providing inspiration to new algorithms.

2 Uncertainty in atomistic diffusion models for structure generation

2.1 Problem definition

As a representative task for machine learning-guided molecule and materials design, we consider generating molecules conditioned on a specific chemical formula. Formally, each atomistic system is described by a tuple $m = (x, F)$, where $x = (x_1, \dots, x_N) \in \mathbb{R}^{N \times 3}$ forms a collection of vectors representing the atom coordinates and $F = (e_1, \dots, e_N) \in \mathbb{Z}^N$ are the associated atom types. Modelling the conditional distribution $p_\theta(x|F) \approx p(x|F)$, rather than the joint distribution $p(x, F)$, provides explicit control over the composition at sampling time. This enables systematic evaluation on both in- and out-of-distribution formulas and prevents it from collapsing to trivial compositions. In what follows, we omit the explicit conditioning on F for brevity.

2.2 Sampling-based generative models

Constructing an expressive model for this conditional distribution requires a flexible generative framework. To this end, we resort to diffusion models [14, 15] that define a generative process by transforming samples from a tractable base distribution into a data distribution $p(x)$ via a learned differential equation. Herein, we adopt the standard diffusion convention where time $t \in [0, 1]$ tracks this evolution, with $t = 0$ corresponding to the data distribution and $t = 1$ to the base distribution.

Forward process Diffusion models define a forward process that gradually perturbs data $x \sim p(x)$ according to the one-step linear transition kernel:

$$p(x_t | x) = \mathcal{N}(x_t | \alpha_t x, \sigma_t^2 \mathbb{I}), \quad (1)$$

where the differentiable schedules α_t and σ_t monotonically decrease and increase, respectively. We use $\alpha_t = 1 - t$ and $\sigma_t = t$. Consequently, noisy samples can be drawn directly from this kernel via the affine transformation $x_t = \alpha_t x + \sigma_t \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \mathbb{I})$.

Equivalently, this kernel corresponds to the conditional distribution of the solution for a linear stochastic differential equation (SDE) initialised at $x_{t=0} = x$:

$$dx_t = f_t x_t dt + g_t dw_t, \quad (2)$$

where w_t denotes the standard Wiener process. The drift and diffusion coefficients, f_t and g_t , are strictly determined by the chosen noise schedules: $f_t = \dot{\alpha}_t / \alpha_t$ and $g_t = \sqrt{2\sigma_t \dot{\sigma}_t - 2\sigma_t^2 \dot{\alpha}_t / \alpha_t}$.

Reverse process To generate data, this forward process must be reversed. Corresponding to the forward SDE described in Eq. (2) is a family of reverse-time SDEs that shares the exact same time-marginals $p_t(x_t)$ [16]:

$$dx_t = \left[f_t x_t - \frac{1 + \eta^2}{2} g_t^2 \nabla_{x_t} \log p_t(x_t) \right] dt + \eta g_t d\hat{w}_t, \quad (3)$$

where $\hat{w}_t$ denotes the reverse-time Wiener process and the scalar $\eta \geq 0$ controls the level of sampling stochasticity. Because the true score governing this process is unknown, diffusion models train a neural network $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ to approximate it.

Training objective The unknown score is generally learned by minimising:

$$\mathcal{L}(\theta) \propto \mathbb{E}_{x, t, x_t} \left[\left\| \nabla_{x_t} \log p(x_t | x) - s_\theta(x_t, t) \right\|_2^2 \right], \quad (4)$$

known as the denoising score matching (DSM) objective. For the Gaussian transition kernel in Eq. (1), the conditional score is analytical: $\nabla_{x_t} \log p(x_t | x) = -(x_t - \alpha_t x) / \sigma_t^2 = -\varepsilon / \sigma_t$. Crucially, while this objective trains the network to match the tractable conditional score, it equates to approximating the true unconditional score up to a constant independent of θ [17].

Parameterisation In practice, as formalised by Li and He [18], mapping the theoretical DSM identity to a practical objective requires two distinct design choices: selecting the output space of the neural network ($y \in \{x, \varepsilon, v, s\}$) and the loss space ($l \in \{x, \varepsilon, v, s\}$). We parameterise our neural network to directly predict the clean data point, $x_\theta \approx x$, while optimising the loss in velocity space, v , where $v = \dot{\alpha}_t x + \dot{\sigma}_t \varepsilon$. In our specific formulation (i.e. $y = x$ and $l = v$), the reconstructed velocity is $v_\theta = \dot{\alpha}_t x_\theta + \dot{\sigma}_t (x_t - \alpha_t x_\theta) / \sigma_t$, and expanding the velocity error yields the following objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{x, t, x_t} \left[\omega_t \|x - x_\theta(x_t, t)\|_2^2 \right], \quad (5)$$

where $\omega_t = (\dot{\alpha}_t - \alpha_t \dot{\sigma}_t / \sigma_t)^2$ is the induced time-weighting on the underlying data prediction error.

2.3 Sampling

Generating novel data requires numerically integrating the reverse SDE in Eq. (3) backwards in time to $t = 0$, initialised from the tractable base distribution $x_1 \sim \mathcal{N}(0, \sigma_1^2 \mathbb{I})$, and using the learned score s_θ . We use an exponential Euler-Maruyama (EM) discretisation scheme [19] as our default numerical integrator (see App. B) and then consider three extensions: EM interleaved with (i) correction steps (PC), (ii) UQ-triggered correction steps (PC-UQ), and (iii) UQ-based look-ahead steps (LA-UQ).

Predictor-corrector PC sampling [15] interleaves unadjusted Langevin (correction) steps between each EM integration (prediction) step. Formally, at time t , the corrector iterates:

$$x_{t, k+1} = x_{t, k} + \delta s_\theta(x_{t, k}, t) + \sqrt{2\delta} \xi, \quad (6)$$

where $\xi \sim \mathcal{N}(0, \mathbb{I})$ and $\delta > 0$ is the step size, which we set dynamically as $\delta = \tau / \mathbb{E} [\|s_\theta(x_t, t)\|^2]$ following Rozet and Louppe [20]. Assuming an exact score and infinitesimal δ , the stationary distribution of this process is p_t , and the corrector therefore naturally contracts x_t back toward the true marginal. Although this does not hold in practice, the corrections often yield higher-quality samples [15, 21]. Our PC-UQ variant applies this mechanism selectively: it triggers Langevin corrections only when the molecule-level uncertainty of x_t (defined in Sec. 2.4) exceeds a threshold S_{thr}^2 , efficiently focusing compute on correcting diverging trajectories.

Look-ahead In LA-UQ, the atom-forcing mechanism of Fig. 1, we adopt the framework of Schusterbauer et al. [13] and aim to reduce the *ambiguity* of *uncertain* atoms during sampling by providing them with a better structural context. LA-UQ relies on diffusion forcing [22], where different dimensions (atoms) can exist at distinct corruption levels, in contrast to standard diffusion models where all dimensions share a single uniform noise level (Sec. 2.2). This enables the network to denoise highly perturbed regions while, in effect, conditioning on cleaner neighbourhoods.

Leveraging this, at time t , we identify *confident* atoms x_t^c via an UQ-quantile threshold q , and propagate them to a cleaner timestep $t' \leq t$ with a multiplier κ . The score for the remaining uncertain atoms x_t^u is then evaluated using this updated context. Formally, the score at each EM step is:

$$\tilde{s}_\theta(x_t, t) = M_c \odot s_\theta(x_t, t) + (1 - M_c) \odot s_\theta(\{x_t^u, x_{t'}^c\}, (1 - M_c) \odot t + M_c \odot t'), \quad (7)$$

where M_c is a binary mask selecting indices of confident atoms. Conditioning x_t^u on the partially denoised state x_t^c , yields more accurate local structures at the cost of one extra forward pass. Because this algorithm evaluates states at mixed noise levels, the score function s_θ strictly requires per-atom time conditioning that enables heterogeneous denoising.

2.4 Uncertainty estimation

Both PC-UQ and LA-UQ require efficient uncertainty estimates during sampling. We model the uncertainty of the predicted velocity field v_θ by assuming a predictive distribution that factorises over atoms. We parameterise the conditional velocity v_a of atom a as an isotropic Gaussian distribution with an input-dependent variance $\sigma_\theta^2(x_t, t)_a$. To learn this variance, we augment the primary objective from Eq. (5) with a weighted negative log-likelihood (NLL) term:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}(\theta) - \lambda \mathbb{E}_{x, t, x_t} \left[\sum_{a=1}^N \sum_{d=1}^3 \log \mathcal{N} \left(v_{a,d} \mid \text{sg}[v_\theta(x_t, t)]_{a,d}, \sigma_\theta^2(x_t, t)_a \right) \right], \quad (8)$$

where the stop-gradient operator $\text{sg}[\cdot]$ prevents the variance objective from interfering with the mean prediction [23]. Full derivations of all UQ-related quantities are provided in App. B.2.

UQ at three levels In the LA-UQ sampler, we directly use the predicted variances $\sigma_\theta^2(x_t, t)$ as the atom-level uncertainty estimates to rank the atoms within the molecule $m_t = (x_t, F)$ at any given time t . For the PC-UQ sampler, we need a coarser signal that decides whether the Langevin corrector should be triggered for m_t at time t . A molecule-level score follows from applying the shared variance assumption at the molecule level instead of the atom level, meaning that the velocities of all atoms in m_t are assumed to have the same variance S^2 . Minimising the negative log-likelihood with this assumption yields

$$S^2(x_t, t)^* = \frac{1}{N} \sum_{a=1}^N \sigma_\theta^2(x_t, t)_a^*, \quad (9)$$

and we therefore define the molecule-level score $S^2(x_t, t)$ as the average of $\sigma_\theta^2(x_t, t)_a$ within m_t . To enable post-hoc filtering, we require an even coarser time-independent molecule-level UQ-score. We obtain this in a similar fashion by simply averaging $S^2(x_t, t)$ over the set of integration steps $\mathcal{T}$

$$S_{\mathcal{T}}^2(\{x_t\}_{t \in \mathcal{T}}) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} S^2(x_t, t), \quad (10)$$

akin to the work of Seij et al. [24].

Normalised UQ-score We hypothesise that ranking atoms directly by σ_θ^2 in LA-UQ and molecules by S^2 in PC-UQ might be problematic. Atoms of different elements and at different t are not on a common scale (see Figs. 11 and 12 in App. D), and the uncertainty estimates are therefore dominated by these systematic offsets rather than the geometric state x_t . To alleviate this, we assume that the expected squared velocity residual $u_a(x_t, t, F) = \frac{1}{3} \mathbb{E}_{x| x_t, t, F} [\|r_a\|^2]$ of atom a (the NLL optimum of $\sigma_\theta^2(x_t, t, F)_a$) factors into an element-conditional baseline that depends only on $(e(a), t, F)$ and a factor that varies with the geometric state x_t , where $r_a = v_\theta(x_t, t, F)_a - v_a$ is the velocity residual, $e(a)$ is the atom type of a , and the dependence on F is again made explicit. That is, we assume

$$u_a(x_t, t, F) = \underbrace{\bar{u}(e(a), t, F)}_{\text{element-conditional baseline}} \cdot \underbrace{\rho_a(x_t, t, F)}_{\text{state-dependent deviation}}, \quad (11)$$

where we define the element-conditional baseline to be the expectation

$$\bar{u}(e, t, F) = \mathbb{E}_{x_t|t, F} [u_a(x_t, t, F)] = \frac{1}{3} \mathbb{E}_{x_t|t, F} [\mathbb{E}_{x| x_t, t, F} [\|r_a\|^2]] \quad (\text{any } a \text{ with } e(a) = e), \quad (12)$$

i.e., the expected residual for an atom of type e in a formula F at time t without knowing the geometry x_t of the molecule². With this definition, the deviation ρ_a becomes the ratio

$$\rho_a(x_t, t, F) = \frac{u_a(x_t, t, F)}{\bar{u}(e(a), t, F)} = \frac{\mathbb{E}_{x| x_t, t, F} [\|r_a\|^2]}{\mathbb{E}_{x_t|t, F} [\mathbb{E}_{x| x_t, t, F} [\|r_a\|^2]]}, \quad (13)$$

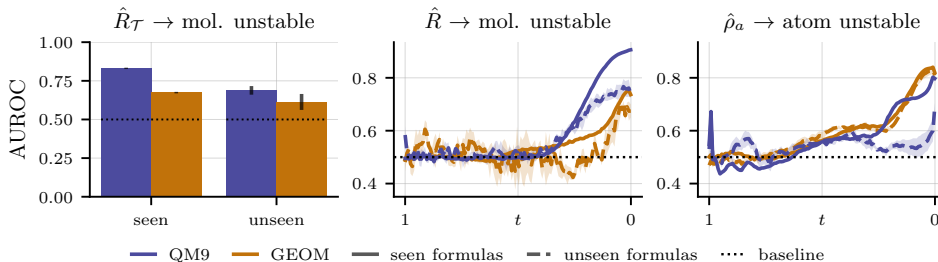


Figure 2: Predictive signal in the normalised UQ-scores from the standard model, measured by AUROC. Molecules are ranked within a formula and atoms within a molecule. **Left:** The post-hoc score $\hat{R}_{\mathcal{T}}$. **Middle:** The molecule score $\hat{R}$ at sampling. **Right:** The atom score $\hat{\rho}_a$ at sampling. Reported AUROC is a per-formula macro average with mean ± 1 standard error over three seeds.

where we have that $\mathbb{E}_{x_t|t,F}[\rho_a(x_t, t, F)] = 1$ and a score $\rho_a > 1$ indicates that an atom’s predicted velocity is more uncertain than expected for its type at the current point in the trajectory. The numerator of ρ_a is what the uncertainty head σ_{θ}^2 estimates. Likewise, we can estimate the denominator by training an auxiliary uncertainty head $\bar{\sigma}_{\theta}^2$ on the same residuals, but with the coarser input $\phi_a = (e(a), t, F)$ that omits all geometry-dependent features. The NLL optimum of $\bar{\sigma}_{\theta}^2$ is the expectation of $\|r_a\|^2$ conditioned on its input. Since ϕ_a fixes t and F but not x_t , we have that

$$\bar{\sigma}_{\theta}^2(\phi_a)^* = \frac{1}{3} \mathbb{E}_{x_t|t,F}[\|r_a\|^2] = \frac{1}{3} \mathbb{E}_{x_t|t,F} \mathbb{E}_{x|t,t,F}[\|r_a\|^2] = \bar{u}(e(a), t, F), \quad (14)$$

i.e., withholding the geometry adds exactly the outer expectation that defines the baseline. The final normalised atom-level UQ-score is therefore the ratio

$$\hat{\rho}_a(x_t, t, F) = \frac{\sigma_{\theta}^2(x_t, t, F)_a}{\bar{\sigma}_{\theta}^2(\phi_a)}, \quad (15)$$

which we use in an alternative LA-UQ-NORM variant of the LA-UQ sampler.

PC-UQ-NORM In PC-UQ, we would like to apply a single global threshold for triggering Langevin correction steps throughout sampling and across formulas, which the raw score S^2 does not support. For this reason, we consider the PC-UQ-NORM sampler, where we aggregate the normalised deviations $\hat{\rho}_a$ rather than the raw variances σ_{θ}^2 , which yields the molecule-level score

$$\hat{R}(x_t, t, F) = \frac{1}{N} \sum_{a=1}^N \hat{\rho}_a(x_t, t, F). \quad (16)$$

By linearity, the corresponding average of the exact deviations, $\frac{1}{N} \sum_{a=1}^N \rho_a(x_t, t, F)$, has expectation 1 for all t and F , simplifying the threshold selection. Finally, we also consider the global normalised molecule-level score $\hat{R}_{\mathcal{T}}$, obtained by substituting Eq. (16) into Eq. (10) in place of S^2 .

3 Related work

Likelihood estimates are notoriously unreliable per-sample confidence signals for generative models [25]. This necessitates alternatives inspired by predictive modelling, where UQ is the standard tool for identifying unreliable outputs. Recently, these techniques have been adapted to sampling-based generative models. Existing approaches broadly fall into two categories depending on whether they aim to model *epistemic* or *aleatoric* uncertainty (or both). Epistemic approaches include computationally efficient diffusion ensembles [26], post-hoc last-layer Laplace approximations [7, 8, 24], Monte-Carlo Dropout [27], or Variational Adaptive Dropout [28]. Lightweight uncertainty heads [29, 10, 13]—the approach we adopt in this work for its computational efficiency—capture total uncertainty, treating it as aleatoric, while purely aleatoric uncertainty can be targeted post-hoc

²The choice of a does not matter as long as $e(a) = e$. The formula distinguishes atoms only by element and our network is permutation equivariant, so two atoms of the same element differ only by the position $x_{t,a}$ that we average over.

Table 1: Test metrics for the different sampling strategies using the standard backbone model. % NFE increase is relative to EM sampling with 125 prediction steps. Results are per-formula macro averages and reported as mean ± 1 standard error across three seeds.

	sampler	steps	% stable		% stable & unique & novel		% NFE increase	
			seen	unseen	seen	unseen	seen	unseen
QM9	EM-125	125	88.21 (40)	83.17 (50)	43.07 (60)	67.49 (87)		
	EM-250	250	88.83 (13)	82.23 (65)	42.44 (94)	65.66 (48)	99.2 (0)	99.2 (0)
	PC	125	94.23 (14)	84.00 (244)	47.81 (69)	68.50 (249)	99.2 (0)	99.2 (0)
	PC-RAND-M	125	81.51 (51)	71.49 (53)	42.86 (74)	61.17 (13)	8.3 (0)	8.2 (1)
	PC-UQ	125	94.76 (16)	85.16 (42)	47.23 (73)	70.09 (66)	84.4 (0)	72.0 (4)
	PC-UQ-M	125	88.59 (16)	81.54 (150)	43.42 (68)	65.84 (67)	5.2 (0)	1.9 (0)
	PC-UQ-NORM	125	95.16 (7)	83.06 (213)	46.25 (101)	67.21 (263)	8.5 (0)	5.3 (0)
	EM-FILTER	125	95.31 (15)	84.71 (73)	40.70 (75)	64.53 (65)	100.0 (0)	100.0 (0)
	EM-FILTER-NORM	125	96.23 (11)	85.40 (50)	40.72 (73)	64.96 (38)	100.0 (0)	100.0 (0)
GEOM	EM-125	125	74.71 (26)	73.14 (12)	74.54 (25)	73.13 (12)		
	EM-250	250	76.66 (14)	74.78 (27)	76.44 (13)	74.78 (27)	99.2 (0)	99.2 (0)
	PC	125	86.88 (14)	86.23 (72)	86.71 (11)	86.23 (72)	99.2 (0)	99.2 (0)
	PC-RAND-M	125	76.40 (24)	75.85 (80)	76.27 (22)	75.85 (80)	12.7 (0)	12.7 (0)
	PC-UQ	125	86.38 (6)	85.88 (45)	86.24 (5)	85.88 (45)	94.2 (0)	93.8 (0)
	PC-UQ-M	125	75.23 (6)	74.49 (20)	75.07 (5)	74.48 (20)	13.1 (0)	12.1 (0)
	PC-UQ-NORM	125	87.33 (11)	86.13 (54)	87.09 (12)	86.13 (54)	12.8 (0)	13.0 (1)
	EM-FILTER	125	80.53 (14)	78.30 (37)	80.27 (13)	78.29 (37)	100.0 (0)	100.0 (0)
	EM-FILTER-NORM	125	81.17 (18)	78.98 (38)	80.90 (19)	78.97 (38)	100.0 (0)	100.0 (0)

via sensitivity analysis [30]; or by exploiting the mathematical structure of the generative process, via the divergence of the vector field or the spread of the posterior distribution of the reverse SDE [31, 32, 11]. In practice, these theoretically distinct quantities are difficult to disentangle; many methods ultimately measure posterior predictive uncertainty, conflating model ignorance, diffusion stochasticity, and data ambiguity [9]. In App. A, we provide a discussion of different approaches to leveraging UQ in diffusion models.

4 Experiments

We conduct a series of experiments to investigate the merits of UQ-based post-hoc filtering, UQ-triggered Langevin corrections, and UQ-based LA sampling for improving sample quality in atomistic diffusion models for structure generation. Additionally, we compare UQ-based samplers using raw and normalised UQ-scores. For full experimental details, we refer to App. C.

Datasets & Model We evaluate on QM9 [33] and GEOM-Drugs [34], restricting the data to molecules with 60 atoms or fewer to reduce computational cost (100% of QM9 and 92.8% of GEOM-Drugs). For a rigorous evaluation, we partition the test sets into *seen* and *unseen* splits based on whether a test molecule’s chemical formula (the generation condition) appeared during training.

The denoiser x_θ and the lightweight uncertainty head $\log \sigma_\theta^2$ are implemented, respectively, as a linear layer and a small MLP applied to a shared PaiNN graph neural network (GNN) backbone [35]. We train two backbone variants: one with standard homogeneous uniform time sampling, and one with multi-dimensional time vectors, sampled via the Logit-Normal Truncated Gaussian (LTG) schedule of Schusterbauer et al. [13], to support the LA samplers’ heterogeneous denoising. The auxiliary baseline uncertainty head, $\log \bar{\sigma}_\theta^2$, is parameterised as a simple GNN operating on embeddings of the chemical formula F and timestep t .

Samplers Our baselines include the EM sampler with 125 (EM-125) and 250 (EM-250) steps, as well as PC, that applies one Langevin correction after each EM-125 step. We compare these against three UQ-augmented methods: (1) EM-FILTER, which generates twice as many candidates per test formula using EM-125 and retains only the 50% most confident; (2) PC-UQ, which selectively triggers corrections on EM-125 trajectories; and (3) LA-UQ, which utilises 125 prediction steps with look-ahead conditioning. We also evaluate the normalised variants of these samplers (denoted with a -NORM suffix), which operate on the normalised scores $\hat{R}_T$, $\hat{R}$, and $\hat{\rho}_a$.

Table 2: Test metrics for the different sampling strategies using the LTG backbone model trained with heterogeneous time sampling. % NFE increase is relative to EM sampling with 125 prediction steps. Results are per-formula macro averages and reported as mean ± 1 standard error across three seeds.

	sampler	steps	% stable		% stable & unique & novel		% NFE increase	
			seen	unseen	seen	unseen	seen	unseen
QM9	EM-125	125	83.23 (56)	74.19 (71)	37.13 (40)	59.66 (79)		
	EM-250	250	82.91 (13)	71.36 (52)	36.54 (69)	58.06 (73)	99.2 (0)	99.2 (0)
	LA-RAND	125	86.90 (64)	73.16 (143)	37.33 (33)	57.28 (91)	93.7 (0)	93.7 (0)
	LA-UQ	125	86.86 (54)	76.81 (59)	37.26 (70)	61.16 (47)	93.7 (0)	93.7 (0)
	LA-UQ-NORM	125	86.12 (68)	75.61 (43)	36.66 (9)	60.80 (28)	93.7 (0)	93.7 (0)
GEOM	EM-125	125	50.80 (15)	49.68 (22)	50.60 (17)	49.67 (22)		
	EM-250	250	52.64 (14)	52.29 (65)	52.32 (15)	52.29 (65)	99.2 (0)	99.2 (0)
	LA-RAND	125	58.49 (28)	56.49 (33)	58.22 (30)	56.49 (33)	93.7 (0)	93.7 (0)
	LA-UQ	125	64.26 (8)	63.12 (14)	63.97 (10)	63.12 (14)	93.7 (0)	93.7 (0)
	LA-UQ-NORM	125	63.31 (28)	62.07 (18)	63.02 (30)	62.07 (18)	93.7 (0)	93.7 (0)

To isolate the specific benefits of the uncertainty estimates, we introduce three control baselines: (1) LA-RAND, which substitutes the UQ-based atomic ranking in LA-UQ with a random ranking; (2) PC-RAND-M, which randomly triggers correction steps at a frequency that matches the computational cost of PC-UQ-NORM; and (3) PC-UQ-M, which uses the standard PC-UQ framework but calibrates its trigger threshold to match the compute cost of PC-UQ-NORM. The PC step scale τ , correction thresholds S_{thr}^2 and $\hat{R}_{\text{thr}}$, and LA parameters q (quantile threshold) and κ (multiplier) were determined with validation set sensitivity analysis (see App. D.8 and Tab. 7).

4.1 Chemical signal of the UQ-scores

We first evaluate whether the normalised UQ-scores $\hat{R}_{\mathcal{T}}$, $\hat{R}$, and $\hat{\rho}_a$, derived from the uncertainty heads σ_{θ}^2 and $\bar{\sigma}_{\theta}^2$ (Sec. 2.4), meaningfully correlate with chemical validity. To do so, we compute the AUROC by treating the UQ-scores as binary classifiers of molecular instability ($\hat{R}_{\mathcal{T}}$ and $\hat{R}$) and atomic instability ($\hat{\rho}_a$). A molecule is defined as stable if the final generated structure maps to a valid SMILES string [36], is not fragmented, and all its constituent atoms are stable. Similarly, an atom is defined as stable if it has the correct valency in the final generated structure [6]. As shown in Fig. 2, the atom-level score $\hat{\rho}_a$ exhibits no predictive signal early in the sampling process (until $t \approx 0.75$) but steadily becomes more informative as generation progresses, except for a late dip on unseen QM9 formulas. The molecule-level score $\hat{R}$ exhibits a similar pattern, albeit emerging later in the process. Finally, the post-hoc trajectory score $\hat{R}_{\mathcal{T}}$ successfully classifies instability, performing better on seen formulas than unseen ones, as expected. Equivalent evaluations using raw UQ-scores exhibit similar trends (Fig. 10 in App. D).

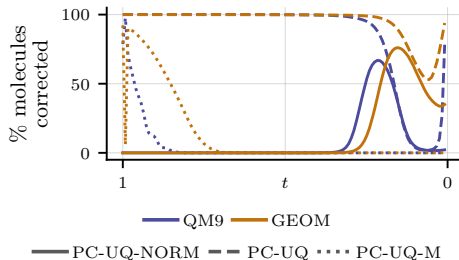


Figure 5: % of molecules corrected at each t with the different UQ-based PC samplers.

4.2 UQ-guided sampling

Having established that the UQ-scores carry meaningful chemical signals, we now investigate whether they can actively improve generation quality. We evaluate each sampler by the percentage of stable molecules generated (Sec. 4.1) and the percentage of stable, unique, and novel (SUN) molecules. Novelty ensures structures are absent from the training data, while uniqueness penalises duplicate generations. We also gauge computational efficiency via the increase in number of function evaluations (NFE) relative to the EM-125 baseline. Note that while % stable and % SUN diverge significantly on QM9, they align closely on GEOM-Drugs due to its vastly larger chemical space.

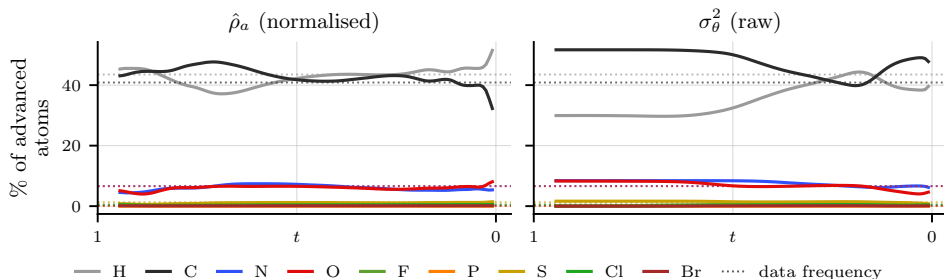


Figure 3: Composition of advanced atoms for the GEOM dataset during LA sampling using the normalised UQ-scores ($\hat{\rho}_a$ in LA-UQ-NORM) and the raw UQ-scores (σ_θ^2 in LA-UQ).

Homogeneous sampling Tab. 1 presents results with the standard homogeneous-time backbone, where all performance differences stem solely from the sampling strategy, as the underlying neural network remains identical. Here EM-FILTER significantly improves stability, corroborating Seij et al. [24], but it severely degrades SUN performance on QM9. Normalising the scores (EM-FILTER-NORM) provides only a marginal edge. In contrast, standard PC sampling strongly improves both stability and SUN. However, UQ variants achieve these gains much more efficiently: PC-UQ reduces the cost, and PC-UQ-NORM matches or even exceeds standard PC performance at a fraction of the NFE overhead. The advantage of normalised UQ-scores becomes even starker at matched compute, where PC-UQ-NORM greatly outperforms the raw-score control, PC-UQ-M.

Heterogeneous LA sampling Next, we evaluate the LA-UQ sampler (Tab. 2), which necessitates the heterogeneous LTG backbone. Comparing Tabs. 1 and 2 reveals a general baseline performance drop caused by the backbone itself. Nonetheless, LA-UQ demonstrates clear utility. On QM9, it yields minor improvements over the EM baselines, outperforming its random control (LA-RAND) strictly on unseen formulas. On the more complex GEOM dataset, LA-UQ substantially outperforms both EM and its random control across all metrics. Interestingly, unlike the PC samplers, using normalised uncertainties slightly degrades LA performance.

4.3 Qualitative sampler behaviour

We analyse the qualitative behaviour of the samplers to understand the mechanisms driving the performance differences between raw and normalised UQ-scores. Fig. 5 reveals stark differences in when the PC variants intervene. PC-UQ applies corrections continuously throughout generation until a late-trajectory dip, whereas the compute-matched PC-UQ-M restricts corrections entirely to the early sampling phase. In contrast, PC-UQ-NORM, guided by the normalised score $\hat{R}$, highly concentrates its correction steps into a narrow, late-stage window of the generative process. Fig. 3 illustrates the elemental composition of the advanced (confident) atoms selected at each integration step by LA-UQ (using raw σ_θ^2) and LA-UQ-NORM (using normalised $\hat{\rho}_a$). As intended, the confident atom composition in LA-UQ-NORM closely tracks the empirical dataset frequencies. Conversely, the raw LA-UQ sampler disproportionately selects Carbon, along with other heavy atoms like Nitrogen and Oxygen, to advance early in the trajectory.

4.4 Examining the UQ-scores

As established in Sec. 2.4, constructing ρ_a to have an expectation of 1 enables comparison of UQ-scores across atom types in the LA-UQ-NORM sampler, and across chemical formulas and timesteps in the PC-UQ-NORM sampler. We now verify whether this property holds in practice by examining if the empirical expectations of the estimated scores, $\hat{\rho}_a$, remain close to 1. Fig. 4 illustrates these empirical expectations for the QM9 dataset (see Fig. 14 in App. D for GEOM), evaluating states x_t drawn from both the forward training process (Eq. (1)) and the reverse sampling process (Eq. (3)). While $\mathbb{E}_{x_t|t,F}[\hat{\rho}_a(x_t, t, F)] \approx 1$ holds faithfully during the forward process, the expectations drift significantly at late t during sampling, revealing a distribution shift in $\hat{\rho}_a$ between training and generation. Theoretically, this shift could pose issues: it risks distorting the atomic rankings for a given t in LA-UQ-NORM, and could be problematic for PC-UQ-NORM if the threshold $\hat{R}_{\text{thr}}$ is not

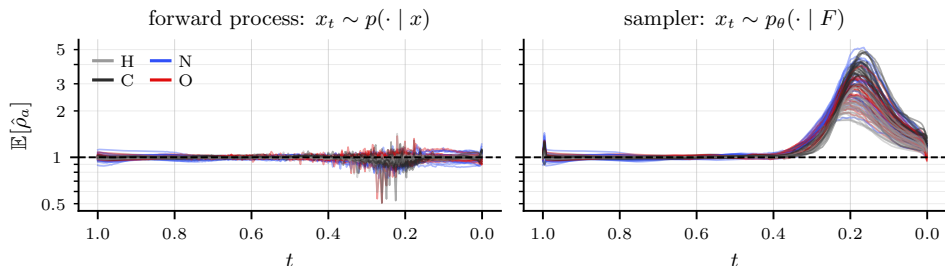


Figure 4: Distribution shift of $\mathbb{E}[\hat{\rho}_a]$ between training and sampling, for the standard model on QM9. Each curve is the empirical expectation of $\hat{\rho}_a$ for one atom type in one chemical formula, against time t , with the states x_t drawn from the forward process (**left**) or from the sampler (**right**). Only test molecules whose formula was seen during training are included.

tuned. However, we empirically observe that PC-UQ-NORM is highly robust, maintaining strong performance across a wide variety of $\hat{R}_{\text{thr}}$ values (Fig. 18 in App. D.8).

5 Discussion

Post-hoc filtering While post-hoc filtering improves chemical validity, in line with previous work, it simultaneously drives SUN below the EM baselines on QM9, indicating that filtering collapses generation toward the training distribution. Consequently, filtering is poorly suited for atomistic diffusion models, where the primary objective is discovering novel drugs and materials.

Predictor-corrector Conversely, Langevin correction steps reliably improve sample quality, and targeting these corrections with UQ achieves these gains at a fraction of the standard computational cost. However, this efficiency stems almost entirely from our UQ-normalisation; naive PC-UQ offers little benefit over standard PC. To our knowledge, PC-UQ-NORM constitutes one of the very first examples of *on-the-fly* active sampling intervention in atomistic diffusion models [28, 27].

Look-ahead sampling Although LA-UQ outperforms the corresponding EM baselines and random controls on GEOM-Drugs, its absolute performance trails the standard homogeneous models (Tab. 1), which we attribute to the underlying LTG backbone model. Although Schusterbauer et al. [13] demonstrated that the heterogeneous time sampling with the LTG schedule outperforms homogeneous uniform time sampling even for EM sampling, the LTG schedule severely undersamples small t (see Fig. 7 in App. D), which might be more detrimental to molecules than images. We also note that the image setting differs fundamentally from our molecular setting: image diffusion can be seen as diffusing node attributes (pixel values) for a fixed geometry (pixel grid), whereas we diffuse the geometry itself (atom positions) with fixed node attributes (atom types). We leave a more in-depth study of heterogeneous denoising for atomistic diffusion models to future work.

Distribution shift The distribution shift observed in Fig. 4 likely arises from either error accumulation or a poor score estimate in the later stage of the reverse process. Either way, this distribution shift complicates calibrating the correction threshold $\hat{R}_{\text{thr}}$ for PC-UQ-NORM. Future work could mitigate this by integrating timestep shifting techniques [37] to realign the training and sampling distributions.

New sampling algorithms Beyond UQ-guided generation, our analysis reveals general sampling dynamics that could be exploited in non-UQ contexts. Specifically, the qualitative behaviours (Sec. 4.3) paired with our generation results suggest two broad strategies: (1) heavily concentrating computational budget in the late stages of generation, and (2) applying asynchronous denoising based on atom type. Notably, resolving Carbon atoms ahead of other elements proved highly effective: a chemically intuitive finding given that Carbon forms the rigid structural backbone of organic molecules, providing a stable scaffold for placing subsequent atoms.

Practicality While we train the variance heads ($\sigma_\theta^2, \bar{\sigma}_\theta^2$) concurrently with the primary denoiser x_θ , these additional components can actually be estimated post-hoc utilising the variance framework of Jordahn et al. [38]. This modularity makes PC-UQ-NORM immediately applicable as a plug-and-play upgrade for existing, large-scale pretrained diffusion models without requiring full retraining.

Conversely, LA samplers lack this immediate portability, as they require heterogeneous denoising backbones, which is not (yet) standard practice in diffusion training.

6 Conclusion

We systematically evaluated how lightweight UQ can actively guide atomistic diffusion models across three granularities. We demonstrate that while post-hoc filtering improves chemical stability, it compromises novelty, making it poorly suited for discovery. Instead, on-the-fly interventions offer a better trade-off. By introducing a normalised UQ-score to remove confounding biases, we enable highly efficient, UQ-triggered Langevin corrections (PC-UQ-NORM). This approach matches standard predictor-corrector performance at a fraction of the computational cost. Finally, while look-ahead sampling highlights the strong potential of asynchronous, atom-level denoising, its immediate utility is bottlenecked by the difficulty of training heterogeneous denoisers for atomistic structures.

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A Related work: leveraging UQ in diffusion models

Uncertainty quantification fits naturally into inference-time scaling [39], where additional compute is leveraged during sampling to improve generation quality. *Filtering* accepts or rejects generations based on trajectory statistics, such as endpoint variance or integrated temporal uncertainty [29, 8, 24, 10, 28]. *Adaptive sampling* acts continuously during the generative process instead of unrolling full trajectories, e.g. by allocating integration steps to uncertain regions [13, 28], by guiding the sampling process away from areas of large uncertainty [30, 10], or by adjusting the variance of the noise injected in the EM sampler based on uncertainty [27]. Finally, *trajectory editing* provides a middle ground, restarting the reverse process at an intermediate timestep (typically where uncertainty is maximised) to re-noise and resample specific dimensions [28].

B Modelling details

B.1 Diffusion model

Size-dependent base distribution In practice, to ensure that molecules of different sizes undergo the same relative corruption, we resort to a base distribution whose variance depends on the number of atoms N ,

$$p(\varepsilon | N) = \mathcal{N}(\varepsilon | 0, \sigma_N^2 \cdot \mathbb{I}), \quad (17)$$

where σ_N is estimated from the training data. It is computed as the standard deviation per coordinate axis for molecules containing N atoms.

For a given molecule size N , we therefore construct noisy samples as

$$x_t = \alpha_t x + \sigma_t \varepsilon, \text{ with } \varepsilon \sim \mathcal{N}(0, \sigma_N^2 \cdot \mathbb{I}).$$

The corresponding effective forward kernel is $p(x_t | x) = \mathcal{N}(x_t | \alpha_t x, \sigma_t^2 \sigma_N^2 \mathbb{I})$, whose score writes,

$$\nabla_{x_t} \log p(x_t | x) = -\frac{x_t - \alpha_t x}{\sigma_t^2 \sigma_N^2}.$$

In practice, we parameterise our neural network as a datapoint predictor. Such that the score is approximated as

$$s_{N,\theta}(x_t, t) = -\frac{x_t - \alpha_t x_\theta(x_t, t)}{\sigma_t^2 \sigma_N^2} = \frac{s_\theta(x_t, t)}{\sigma_N^2}.$$

Predictor step By default, we use a stochastic exponential-Euler integrator (we denote it EM throughout) and decompose the drift into a semilinear form: a linear term integrated exactly and a non-linear term involving the neural network prediction integrated using a left-point rule. For a molecule with N atoms, we integrate Eq. (3) from s to t (with $t \leq s$), as

$$x_t = \frac{\alpha_t}{\alpha_s} x_s + (1 + \eta^2) \sigma_s^2 \left(\frac{\alpha_t}{\alpha_s} - \frac{\sigma_t}{\sigma_s} \right) s_\theta(x_s, s) + \eta \sigma_t \sqrt{1 - \left(\frac{\sigma_t / \sigma_s}{\alpha_t / \alpha_s} \right)^2} \xi_N, \quad (18)$$

where $\xi_N \sim \mathcal{N}(0, \sigma_N^2 \cdot \mathbb{I})$. For stochastic sampling, we use $\eta = 1$, where $\eta = 0$ corresponds to the deterministic case.

Corrector step A common technique to improve sampling quality in diffusion models is to interleave standard predictor steps that integrate the reverse SDE, as that of Eq. (18), for instance, with so-called corrector steps that operate at a fixed corruption level. A standard corrector step usually amounts to performing a fixed number of unadjusted Langevin steps starting from the output of the predictor, $x_{t,0} = x_t$,

$$x_{t,k+1} = x_{t,k} + \delta \cdot s_{N,\theta}(x_{t,k}, t) + \sqrt{2\delta} \cdot \xi, \quad (19)$$

with $\xi \sim \mathcal{N}(0, \mathbb{I})$. Following previous work [20], we seek to set the step size δ dynamically, based on the expected magnitude of the score,

$$\delta = \frac{\tau}{\mathbb{E}[\|s_{N,\theta}(x_t, t)\|^2]}.$$

B.2 Uncertainty estimation

To model the uncertainty of the conditional velocity $v(x, x_t, t)$, we use the following likelihood

$$\begin{aligned}
p(v \mid x_t, t, F) &= \prod_{a=1}^N p(v_a \mid x_t, t, F) && \text{(cond. indep. per-atom velocities)} \\
&= \prod_{a=1}^N \prod_{d=1}^3 p(v_{a,d} \mid x_t, t, F) && \text{(cond. indep. coordinates)} \\
&= \prod_{a=1}^N \prod_{d=1}^3 \mathcal{N}(v_{a,d} \mid v_\theta(x_t, t, F)_{a,d}, \sigma_\theta^2(x_t, t, F)_a), && (20)
\end{aligned}$$

where (x_t, t, F) is the network input. We note that a single variance $\sigma_\theta^2(x_t, t, F)_a$ is emitted across all dimensions d for the velocity of atom a , i.e., the velocity v_a is assumed to have an isotropic Gaussian distribution. The negative log-likelihood (up to an additive constant) for a single training point (x, x_t, F) at time t is therefore

$$\begin{aligned}
\ell(x, x_t, t, F) &= \sum_{a=1}^N \sum_{d=1}^3 \frac{1}{2} \left(\frac{r_{a,d}^2}{\sigma_\theta^2(x_t, t, F)_a} + \log \sigma_\theta^2(x_t, t, F)_a \right), \quad r_{a,d} = v_\theta(x_t, t, F)_{a,d} - v_{a,d} \\
&= \sum_{a=1}^N \left(\frac{\|r_a\|^2}{2\sigma_\theta^2(x_t, t, F)_a} + \frac{3}{2} \log \sigma_\theta^2(x_t, t, F)_a \right), && (21)
\end{aligned}$$

and the NLL term of Eq. (8) is

$$\mathcal{L}_{\text{NLL}}(\theta) = \mathbb{E}_{x, x_t, t, F}[\ell(x, x_t, t, F)] = \mathbb{E}_F \mathbb{E}_{x|F} \mathbb{E}_t \mathbb{E}_{x_t|x, t}[\ell(x, x_t, t, F)]. \quad (22)$$

Optimum of the uncertainty head and its decomposition The optimum of the uncertainty head for an input (x_t, t, F) is

$$\sigma_\theta^2(x_t, t, F)_a^* = \frac{1}{3} \mathbb{E}_{x|x_t, t, F}[\|r_a\|^2]. \quad (23)$$

We can see this by rearranging the expectation in $\mathcal{L}_{\text{NLL}}(\theta)$

$$\mathcal{L}_{\text{NLL}}(\theta) = \mathbb{E}_{x_t, t, F} \mathbb{E}_{x|x_t, t, F}[\ell(x, x_t, t, F)] \quad (24)$$

$$= \mathbb{E}_{x_t, t, F} \left[\sum_{a=1}^N \left(\frac{\mathbb{E}_{x|x_t, t, F}[\|r_a\|^2]}{2\sigma_\theta^2(x_t, t, F)_a} + \frac{3}{2} \log \sigma_\theta^2(x_t, t, F)_a \right) \right] \quad (25)$$

and realising that we can minimise $\mathcal{L}_{\text{NLL}}(\theta)$ by minimising the integrand of the outer expectation $\mathbb{E}_{x|x_t, t, F}[\cdot]$ pointwise for each input. Thus for a given network input (x_t, t, F) , fixed $v_\theta(x_t, t, F)$, and a sufficient capacity of the uncertainty head $\sigma_\theta^2(x_t, t, F)$, the optimum of the uncertainty head is obtained by solving

$$\frac{d\mathbb{E}_{x|x_t, t, F}[\ell(x, x_t, t, F)]}{d\sigma_\theta^2(x_t, t, F)_a} = \frac{3}{2\sigma_\theta^2(x_t, t, F)_a} - \frac{\mathbb{E}_{x|x_t, t, F}[\|r_a\|^2]}{2\sigma_\theta^4(x_t, t, F)_a} = 0 \quad (26)$$

for $\sigma_\theta^2(x_t, t, F)_a$, which yields the optimum in Eq. (23). In practice, the uncertainty head σ_θ^2 only approximately attains this optimum. Since the NLL term is down-weighted by λ in the loss, the model backbone receives a correspondingly weaker gradient signal from it, and representation capacity is expected to be primarily allocated to the mean head v_θ .

The optimum of the uncertainty head can be decomposed into a model error term and an ambiguity term, which is seen by adding and subtracting the marginal velocity $\bar{v}_a = \mathbb{E}_{x|x_t, t, F}[v_a]$ (that $v_\theta(x_t, t, F)_a$ estimates) from the residual r_a and by applying the identity $\mathbb{E}[\|y\|^2] = \|\mathbb{E}[y]\|^2 + \text{tr}(\mathbb{C}[y])$. That is

$$r_a = v_\theta(x_t, t, F)_a - v_a = (v_\theta(x_t, t, F)_a - \bar{v}_a) + (\bar{v}_a - v_a), \quad (27)$$

which gives us

$$\mathbb{E}_{x|x_t, t, F}[r_a] = v_\theta(x_t, t, F)_a - \bar{v}_a \quad (28)$$

$$\mathbb{C}_{x|x_t, t, F}[r_a] = \mathbb{C}_{x|x_t, t, F}[v_a], \quad (29)$$

and therefore

$$\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2] = \underbrace{\|v_\theta(x_t, t, F)_a - \bar{v}_a\|^2}_{\text{model error}} + \underbrace{\text{tr}(\mathbb{C}_{x|x_t,t,F}[v_a])}_{\substack{\text{ambiguity} \\ (\text{irreducible given } x_t, t, F)}}. \quad (30)$$

Thus, the predicted uncertainty contains both the error of the estimated velocity v_θ and the posterior spread of the conditional velocity v_a at (x_t, t, F) , i.e., its ambiguity. The model error can be reduced by, e.g., collecting more data or increasing model capacity. In contrast, $\mathbb{C}_{x|x_t,t,F}[v_a]$ represents a genuine ambiguity: there are potentially many clean structures x consistent with the noisy state x_t , and they imply different velocities v_a for atom a .

B.2.1 Atom-level uncertainty signal

In LA sampling, we wish to rank atoms within a molecule based on their uncertainty and advance the most confident atoms to a cleaner state such that their resolved positions can provide a more informative context for the uncertain atoms. This can be seen as an attempt to reduce the ambiguity term in Eq. (30) by replacing the context (x_t, t, F) with an altered context (x'_t, t, F) , where $x'_t = \{x_t^u, x_t^c\}$ and x^u and x^c denote the set of uncertain and confident atoms, respectively, and t' is the look-ahead time of Sec. 2.3.

However, ranking atoms directly by σ_θ^2 is problematic: atoms of different elements are not on a common scale (see Figs. 11 and 12), and the resulting ranking is dominated by this systematic offset rather than the current geometric state x_t . We therefore assume that the expected squared velocity residual $u_a(x_t, t, F) = \frac{1}{3}\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2]$ of atom a (the NLL optimum of $\sigma_\theta^2(x_t, t, F)_a$) factors into an element-conditional baseline that depends only on $(e(a), t, F)$, where $e(a)$ is the atom type of a , and a factor that varies with the geometric state x_t , i.e.,

$$u_a(x_t, t, F) = \underbrace{\bar{u}(e(a), t, F)}_{\substack{\text{element-conditional} \\ \text{baseline}}} \cdot \underbrace{\rho_a(x_t, t, F)}_{\substack{\text{state-dependent} \\ \text{deviation}}}, \quad (31)$$

and rank atoms based on the state-dependent deviation $\rho_a(x_t, t, F)$. We define the element-conditional baseline to be the expectation

$$\bar{u}(e, t, F) = \mathbb{E}_{x_t|t,F}[u_a(x_t, t, F)] = \frac{1}{3}\mathbb{E}_{x_t|t,F}[\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2]] \quad (\text{any } a \text{ with } e(a) = e), \quad (32)$$

i.e., the expected velocity residual for an atom of type e in a formula F at time t without knowing the geometry x_t of the molecule³. With this definition, the state-dependent deviation $\rho_a(x_t, t, F)$ becomes the ratio

$$\rho_a(x_t, t, F) = \frac{u_a(x_t, t, F)}{\bar{u}(e(a), t, F)} = \frac{\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2]}{\mathbb{E}_{x_t|t,F}[\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2]]}, \quad (33)$$

where we have that $\mathbb{E}_{x_t|t,F}[\rho_a(x_t, t, F)] = 1$ and a score $\rho_a(x_t, t, F) > 1$ indicates that an atom's predicted velocity is more uncertain than expected for its type at the current point in the trajectory. We note that the ratio allows comparison not only across atom types but also across formulas and timesteps, which will become relevant in App. B.2.2.

The numerator of ρ_a is exactly what the uncertainty head σ_θ^2 estimates, while the denominator is an intractable expectation. The denominator can be estimated with, e.g., Monte Carlo integration, but that would render sampling prohibitively expensive. Instead, we estimate it by training an auxiliary uncertainty head $\bar{\sigma}_\theta^2$ on the same residuals, but with the coarser input $\phi_a = (e(a), t, F)$ that omits all geometry-dependent features.

The optimum of $\bar{\sigma}_\theta^2$ can be derived analogously to σ_θ^2 in App. B.2, i.e., its optimum is the expectation of $\|r_a\|^2$ conditioned on its input. Since ϕ_a fixes t and F but not x_t , we have that

$$\bar{\sigma}_\theta^2(\phi_a)^* = \frac{1}{3}\mathbb{E}_{x,x_t|t,F}[\|r_a\|^2] = \frac{1}{3}\mathbb{E}_{x_t|t,F}\mathbb{E}_{x|x_t,t,F}[\|r_a\|^2] = \bar{u}(e(a), t, F), \quad (34)$$

³The choice of a does not matter as long as $e(a) = e$. The formula distinguishes atoms only by their element, and our network is equivariant to atom permutations, so two atoms of the same element differ only by their position $x_{t,a}$, which is what we average over.

i.e., withholding the geometry adds exactly the outer expectation that defines the baseline. The final ranking of atoms by uncertainty is therefore done by the ratio

$$\hat{\rho}_a(x_t, t, F) = \frac{\sigma_{\theta}^2(x_t, t, F)_a}{\bar{\sigma}_{\theta}^2(\phi_a)}. \quad (35)$$

While this ratio could become unstable if $\bar{\sigma}_{\theta}^2(\phi_a) \rightarrow 0$, we empirically never observe this to be the case, as the residuals of the predicted vector field never approach 0 during training.

B.2.2 Time-varying molecule-level uncertainty signal

In the UQ-based PC samplers, we need a single score per molecule that decides whether to trigger a Langevin correction step for the molecule at the current state. Like at the atom level, we aim to target molecules that are more *unusual* than expected at the current state, rather than molecules that are uncertain in absolute terms. Furthermore, the score should be comparable across formulas and timesteps to allow for a single threshold value.

From atoms to molecules A molecule-level uncertainty score follows from the same derivation as in App. B.2, but where the shared variance assumption is applied at a coarser level. That is, assuming that the velocity of all atoms in the molecule is modelled with a single shared variance S^2 , $p(v \mid x_t, t, F) = \prod_{a=1}^N \prod_{d=1}^3 \mathcal{N}(v_{a,d} \mid v_{\theta}(x_t, t, F)_{a,d}, S^2)$, then minimising the negative log-likelihood gives

$$S^2(x_t, t, F)^* = \frac{1}{3N} \mathbb{E}_{x \mid x_t, t, F} [\|r\|^2] = \frac{1}{3N} \mathbb{E}_{x \mid x_t, t, F} \left[\sum_{a=1}^N \|r_a\|^2 \right] = \frac{1}{N} \sum_{a=1}^N \sigma_{\theta}^2(x_t, t, F)_a^*. \quad (36)$$

Thus we can obtain a molecule-level uncertainty score by simply averaging our predicted atom-level variances.

Normalisation However, directly averaging the raw atom-level predictions would inherit the same scale confounds as in App. B.2.1 and the scale of the uncertainty would depend on the chemical formula F (and implicitly molecule size) and the timestep t . For this reason, we aggregate the normalised deviations rather than raw variances,

$$\hat{R}(x_t, t, F) = \frac{1}{N} \sum_{a=1}^N \hat{\rho}_a(x_t, t, F). \quad (37)$$

By linearity, the corresponding average of the exact deviations, $\frac{1}{N} \sum_{a=1}^N \rho_a(x_t, t, F)$, has expectation 1 for every formula and every timestep, which allows us to apply a single threshold $\hat{R}_{\text{thr}}$ for triggering Langevin correction steps throughout sampling and across formulas.

C Experimental details

Datasets We use QM9 [33] and GEOM-Drugs [34]. For GEOM-Drugs, we keep the first conformer of each molecule, discard molecules with more than 60 atoms, and keep the dataset’s standard random split, which already leaves many test formulas unseen. For QM9, we re-split the data by chemical formula, so that a formula generally only exists in one data split: each formula is one allocation group, formulas with more than 250 isomers are cut into groups of at most 250 and are therefore shared between splits, and groups are allocated so that the atom-count distribution is matched across splits. Tab. 3 gives the resulting sizes. The *seen* and *unseen* columns of our results tables split the test set by whether a molecule’s formula occurs in the training set.

Table 3: Dataset splits in number of molecules. *Unseen* counts the test formulas that do not occur in the training set, with the number of corresponding test molecules in parentheses.

	train	val	test	test formulas	unseen
QM9	99,914	17,528	13,389	89	53 (2,242)
GEOM-Drugs	225,734	28,236	28,229	15,771	3,101 (3,246)

Model Both backbones are a PaiNN-style equivariant GNN [35] with 8 layers and hidden dimension 256, applied to the fully connected molecular graph (about 15 M parameters). We keep PaiNN’s message and update blocks and add DIT-style adaptive (equivariant) layer normalisation [40] conditioned on the timestep. Messages are gated by a sigmoid of the message itself [41] and by the two endpoints’ timesteps, so a message depends on the noise level of both of its atoms. Each atom enters as an atom-type embedding plus a molecule-size embedding, concatenated with a timestep embedding. The denoiser head x_θ is a linear readout of the final vector features, and the uncertainty head $\log \sigma_\theta^2$ a two-layer MLP on the final scalar features and vector norms. The baseline head $\log \bar{\sigma}_\theta^2$ is a three-layer MLP that reads an atom’s input embedding concatenated with the mean of those embeddings over the molecule (in essence a very simple GNN), so it sees the atom type, timestep and formula but no geometry, as App. B.2.1 requires.

Training We minimise Eq. (8): the velocity-space objective plus the NLL term of each uncertainty head, both weighted by $\lambda = 0.01$ and both with a stop-gradient on the mean. Additionally, we detach the input to the baseline uncertainty head $\bar{\sigma}_\theta^2$. We use AdamW with a learning rate of $2 \cdot 10^{-4}$, batch size 64, gradient clipping at norm 1, and an exponential moving average of the weights. The standard backbone draws one timestep per molecule, stratified uniformly across the batch, the LTG backbone draws one per atom from the LTG schedule of Schusterbauer et al. [13], with a logit-normal location of 0.7, a scale of 1.0, and a per-atom standard deviation of 0.1. Each model was trained on a single GPU: 1000 epochs (about 3 days) on QM9, and 300 epochs (about 7 days) on GEOM-Drugs. Every 5 epochs we sample 2048 validation molecules with 250 EM steps, and keep the checkpoint with the highest % stable.

Evaluation Each evaluation run generates one structure per test molecule, conditioned on that molecule’s formula, so a generated set has the same size and formula composition as the test set. Every configuration is run with three seeds, and we report the mean ± 1 standard error over them. A molecule counts as stable if it maps to a valid SMILES string [36], forms a single fragment, and all of its atoms have a valid valency; on QM9 bond orders are read off an interatomic-distance table, and on GEOM-Drugs bonds are perceived with RDKit [42] and the valency check follows Nikitin et al. [6]. It counts as SUN if it is also unique within the generated set and absent from the training data. The main tables average per formula, that is, each metric is computed within a formula and formulas are then weighted equally. NFE is the average number of network calls per molecule *delivered*, measured during sampling, and is reported relative to the 126 calls of EM-125.

Samplers All samplers use 125 prediction steps, except EM-250. After each prediction step, the PC samplers apply a Langevin correction step (Eq. (6)) to the molecules whose trigger fires: PC fires on every molecule, PC-UQ, PC-UQ-M and PC-UQ-NORM on those whose UQ-score exceeds their threshold, and PC-RAND-M with a fixed probability. Each correction costs one extra network call for that molecule alone, and Fig. 5 shows when the triggers fire. The LA samplers rank the atoms within each molecule, by σ_θ^2 (LA-UQ), by $\hat{\rho}_a$ (LA-UQ-NORM) or at random (LA-RAND), advance the most confident q -fraction of them to $t' = 1 - \min(\kappa(1 - t), 1)$, and re-evaluate the score of the remaining atoms in that context. The extra network call this needs is skipped while $t \geq 0.95$, so the LA samplers cost 93.7% rather than 99.2% more than EM-125. EM-FILTER generates the test set twice and keeps, within each formula, half of the candidates with the lowest trajectory-averaged score, delivering as many molecules as a single plain run at twice the cost.

Hyperparameters The number of prediction steps B , the corrector step size τ , the two correction thresholds $\hat{R}_{\text{thr}}$ and S_{thr}^2 , and the LA parameters q and κ were each chosen by a separate sweep on the full validation set with three seeds. Tab. 4 lists the grids, App. D.8 shows the sweeps themselves, and Tab. 7 gives the chosen values. The two compute-matched controls are matched rather than swept: PC-RAND-M fires as often as PC-UQ-NORM was measured to fire on validation, and PC-UQ-M takes the threshold whose validation NFE is closest to PC-UQ-NORM’s. The LA parameters q and κ were swept using LA-UQ-NORM and reused for LA-UQ and LA-RAND.

Table 4: The validation sweeps behind Tab. 7. Each parameter was swept on its own, on the full validation split with three seeds: B on both backbones, τ and the two thresholds on the standard backbone, and (q, κ) jointly on the LTG backbone.

parameter	sampler	data	grid
B	EM	QM9 GEOM	25, 50, 75, ..., 250 50, 75, 100, ..., 250
τ	PC	QM9 GEOM	0.01, 0.025, 0.05, 0.075, 0.1, 0.15, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10.0 0.05, 0.1, 0.15, 0.25, 0.5, 0.75, 1.0, 1.5, 2.5, 3.0, 4.0, 5.0
$\hat{R}_{\text{thr}}$	PC-UQ-NORM	both	1.0, 1.025, 1.05, 1.1, 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, 1.75, 2.0, 2.25, 2.5, 3.0
S_{thr}^2	PC-UQ	QM9 GEOM	15 log-spaced values from 0.068 to 2.99 15 log-spaced values from 0.092 to 7.88
q, κ	LA-UQ-NORM	both	all 16 combinations of $q \in \{0.2, 0.4, 0.6, 0.8\}$ and $\kappa \in \{1.2, 1.4, 1.6, 1.8\}$

D Additional results

D.1 Test results as per-molecule averages

Reading the results tables The parenthesised values are the standard errors in the last displayed digits, so 88.21 (40) is 88.21 ± 0.40 . Bold marks the best value in a column within a dataset block. NFE counts model calls per molecule *delivered*, which is why the filtering samplers cost +100% as it generates twice and keeps half.

Table 5: Test metrics for the different sampling strategies using the standard backbone model. % NFE increase is relative to EM sampling with 125 prediction steps. Results are per-molecule averages and reported as mean ± 1 standard error across three seeds.

	sampler	steps	% stable		% stable & unique & novel		% NFE increase	
			seen	unseen	seen	unseen	seen	unseen
QM9	EM-125	125	88.77 (16)	83.90 (60)	43.23 (23)	64.73 (56)		
	EM-250	250	89.46 (10)	84.83 (20)	42.80 (40)	64.73 (38)	99.2 (0)	99.2 (0)
	PC	125	94.77 (8)	86.40 (45)	47.19 (12)	66.31 (68)	99.2 (0)	99.2 (0)
	PC-RAND-M	125	81.90 (39)	76.94 (40)	42.48 (26)	61.83 (61)	8.3 (0)	8.3 (0)
	PC-UQ	125	95.15 (13)	87.85 (40)	47.34 (41)	65.45 (33)	84.6 (0)	80.4 (0)
	PC-UQ-M	125	88.78 (23)	83.54 (14)	43.39 (38)	63.83 (16)	5.0 (0)	3.8 (0)
	PC-UQ-NORM	125	95.60 (9)	88.06 (58)	45.69 (17)	66.49 (63)	8.6 (0)	6.9 (0)
	EM-FILTER	125	95.97 (5)	87.53 (32)	41.17 (14)	62.55 (49)	100.0 (0)	100.0 (0)
	EM-FILTER-NORM	125	96.85 (4)	88.54 (19)	41.22 (16)	62.56 (43)	100.0 (0)	100.0 (0)
GEOM	EM-125	125	75.12 (19)	73.23 (14)	74.93 (17)	73.21 (13)		
	EM-250	250	76.96 (20)	74.80 (25)	76.74 (19)	74.80 (25)	99.2 (0)	99.2 (0)
	PC	125	87.03 (12)	86.22 (69)	86.85 (10)	86.22 (69)	99.2 (0)	99.2 (0)
	PC-RAND-M	125	77.00 (7)	75.90 (78)	76.86 (8)	75.89 (79)	12.7 (0)	12.7 (0)
	PC-UQ	125	86.63 (8)	85.97 (46)	86.49 (7)	85.97 (46)	94.3 (0)	93.8 (1)
	PC-UQ-M	125	75.51 (9)	74.52 (13)	75.33 (8)	74.51 (13)	13.8 (0)	12.1 (0)
	PC-UQ-NORM	125	87.51 (7)	86.20 (52)	87.26 (7)	86.20 (52)	12.7 (0)	13.0 (1)
	EM-FILTER	125	81.49 (15)	78.31 (36)	81.20 (14)	78.29 (35)	100.0 (0)	100.0 (0)
	EM-FILTER-NORM	125	82.18 (19)	78.97 (32)	81.88 (19)	78.95 (32)	100.0 (0)	100.0 (0)

Table 6: Test metrics for the different sampling strategies using the LTG backbone model trained with heterogeneous time sampling. % NFE increase is relative to EM sampling with 125 prediction steps. Results are per-molecule averages and reported as mean ± 1 standard error across three seeds.

	sampler	steps	% stable		% stable & unique & novel		% NFE increase	
			seen	unseen	seen	unseen	seen	unseen
QM9	EM-125	125	84.47 (35)	76.93 (33)	38.19 (8)	57.14 (38)		
	EM-250	250	83.55 (11)	74.29 (19)	37.32 (17)	55.65 (21)	99.2 (0)	99.2 (0)
	LA-RAND	125	88.25 (17)	78.00 (37)	38.66 (35)	56.85 (21)	93.7 (0)	93.7 (0)
	LA-UQ	125	88.13 (11)	78.90 (55)	38.01 (10)	57.64 (86)	93.7 (0)	93.7 (0)
	LA-UQ-NORM	125	88.10 (16)	78.31 (58)	38.34 (7)	57.36 (60)	93.7 (0)	93.7 (0)
GEOM	EM-125	125	50.99 (18)	49.67 (26)	50.77 (19)	49.66 (26)		
	EM-250	250	53.13 (12)	52.22 (69)	52.79 (12)	52.22 (69)	99.2 (0)	99.2 (0)
	LA-RAND	125	58.66 (6)	56.70 (37)	58.36 (7)	56.70 (37)	93.7 (0)	93.7 (0)
	LA-UQ	125	64.59 (1)	63.04 (12)	64.23 (2)	63.04 (12)	93.7 (0)	93.7 (0)
	LA-UQ-NORM	125	63.76 (23)	62.10 (10)	63.40 (23)	62.10 (10)	93.7 (0)	93.7 (0)

D.2 Molecule scores during sampling

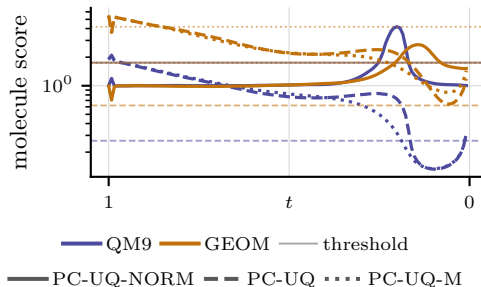


Figure 6: Median of the molecule scores $\hat{R}$ (PC-UQ-NORM) and S^2 (PC-UQ and PC-UQ-M) during sampling with the UQ-based PC samplers.

D.3 Marginal time distributions

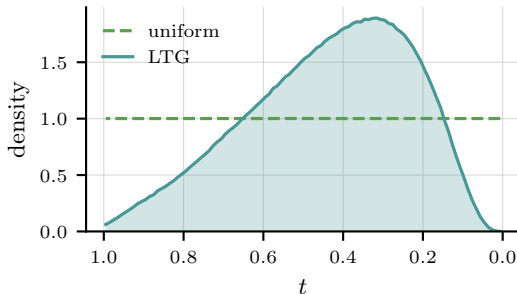


Figure 7: Marginal distribution of t with standard homogeneous uniform time sampling and with heterogeneous LTG time sampling

D.4 Composition of advanced atoms for QM9

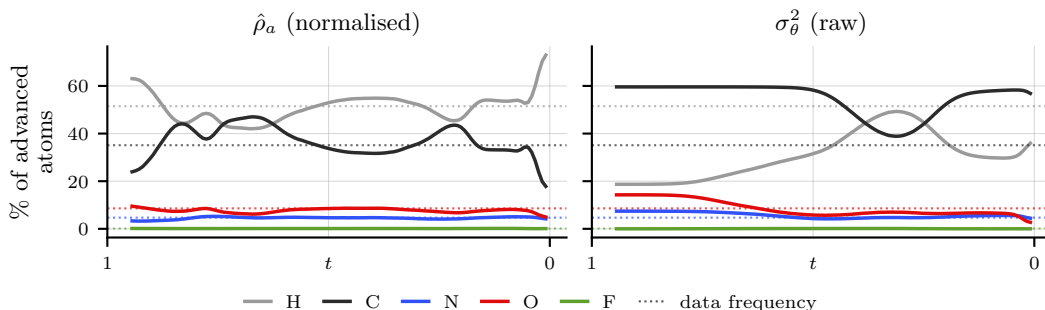


Figure 8: Composition of advanced atoms for the QM9 dataset during LA sampling using the normalised UQ-scores ($\hat{\rho}_a$ in LA-UQ-NORM) and the raw UQ-scores (σ_θ^2 in LA-UQ).

D.5 Uncertainty signal diagnostics

Coverage of the diagnostics figures A within-group rank metric needs a group holding at least two atoms/molecules of both outcomes (stable/unstable), so Figs. 2, 9 and 10 are measured on the formulas (for $\hat{R}_T$ and $\hat{R}$) and the molecules (for $\hat{\rho}_a$) that satisfy this criterion. This covers essentially all of QM9’s seen split but only 42% of GEOM’s and $\sim 3\%$ of GEOM’s unseen split (about 100 molecules), and the atom-level panel is restricted to molecules ending with both stable and unstable atoms (7–15% of atoms on the standard backbone).

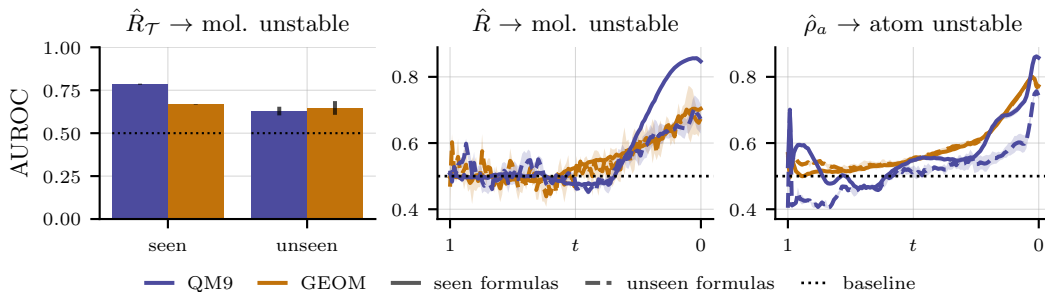


Figure 9: Assessment of the predictive signal in the uncertainty scores from the LTG backbone model trained with heterogeneous time sampling. **Left:** Within-formula AUROC of predicting molecule instability using the time-independent molecule-level score $\hat{R}_T$ after sampling. **Middle:** Within-formula AUROC of predicting molecule instability using the time-varying molecule-level score $\hat{R}$ during sampling. **Right:** Within-molecule AUROC of predicting atom instability using the time-varying atom-level score $\hat{\rho}_a$ during sampling. AUROCs are computed as per-formula macro averages and reported as mean ± 1 standard error across three seeds.

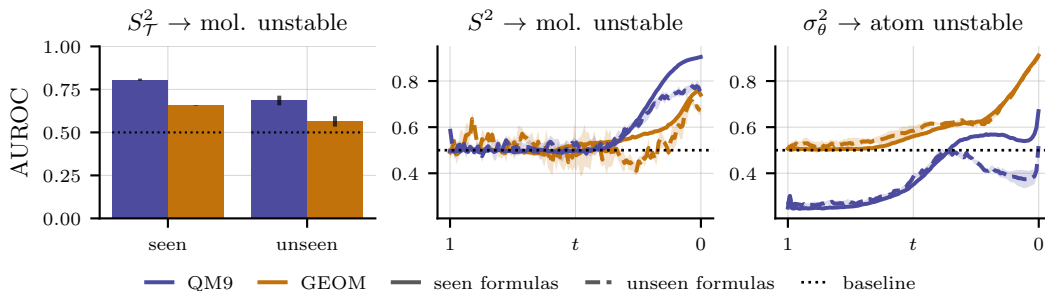


Figure 10: Assessment of the predictive signal in the raw (unnormalised) uncertainty scores from the standard backbone model. **Left:** Within-formula AUROC of predicting molecule instability using the time-independent molecule-level score S_T^2 after sampling. **Middle:** Within-formula AUROC of predicting molecule instability using the time-varying molecule-level score S^2 during sampling. **Right:** Within-molecule AUROC of predicting atom instability using the time-varying atom-level score $\sigma_\theta^2(x_t, t)_a$ during sampling. AUROCs are computed as per-formula macro averages and reported as mean ± 1 standard error across three seeds.

D.6 Uncertainty-scale per element

Coverage of the uncertainty-scale figures Elements with fewer than 50 atoms are not drawn in Figs. 11 and 12 (boron on GEOM-Drugs). Iodine and phosphorus only have 54 and 324 atoms pooled over three seeds, so their bands carry far less evidence than the others.

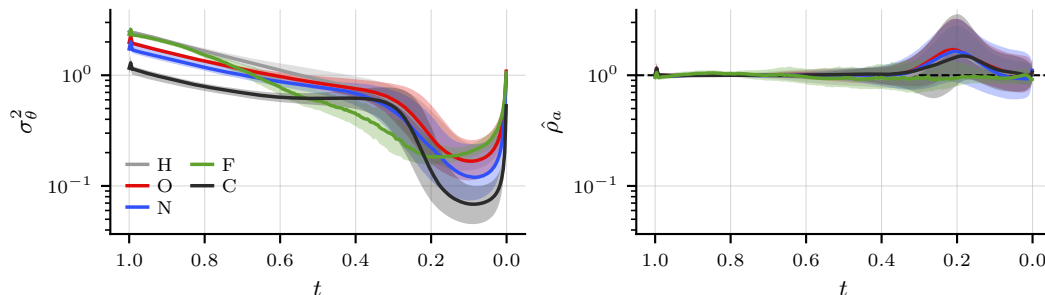


Figure 11: Median and interquartile range per element of the uncertainty scores from the standard backbone model on the QM9 dataset. **Left:** The raw UQ-score $\sigma_\theta^2(x_t, t, F)_a$. **Right:** The normalised UQ-score $\hat{\rho}_a$.

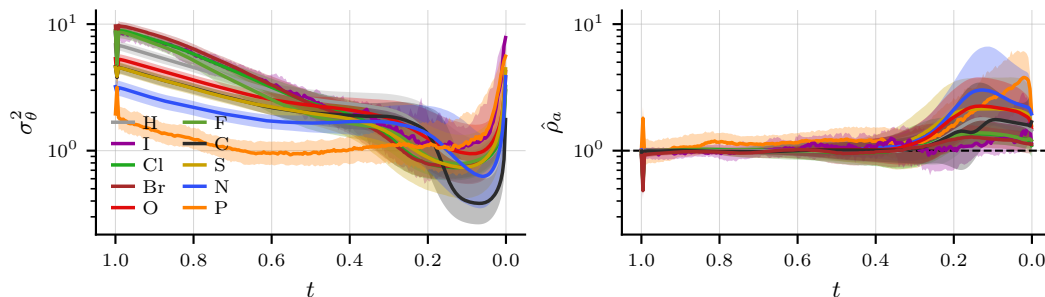


Figure 12: Median and interquartile range per element of the uncertainty scores from the standard backbone model on the GEOM-Drugs dataset. **Left:** The raw UQ-score $\sigma_\theta^2(x_t, t, F)_a$. **Right:** The normalised UQ-score $\hat{\rho}_a$.

D.7 Distribution shift of the atom-level uncertainty signal

Coverage of the distribution shift figures Each curve in Figs. 4 and 13 to 15 averages $\hat{\rho}_a$ over the atoms of one element within one chemical formula. We keep a formula only if at least 8 of its molecules are available both in the forward-noised set (left panel) and in the sampled set (right panel). On QM9, 35 of the 36 seen formulas are kept and cover all but two of the seen test molecules. On GEOM-Drugs, only 345 of the 12,670 seen formulas are kept, covering 14% of the seen test molecules, and they contain only H, C, N, O and S, since no formula holding a halogen or phosphorus has enough molecules to qualify. The GEOM panels therefore rest on a seventh of their split, whereas the QM9 panels rest on all of theirs.

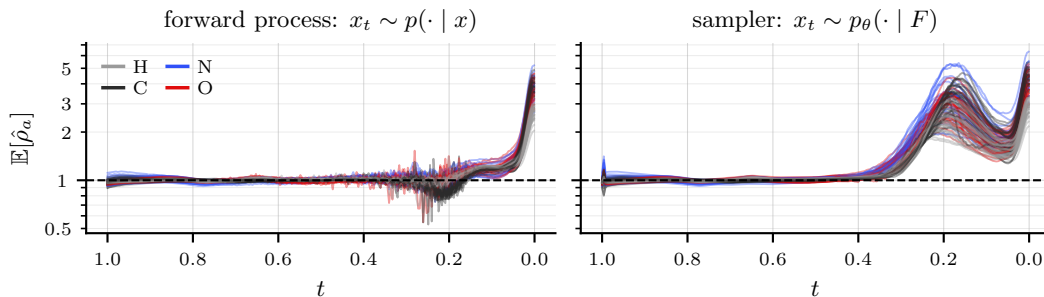


Figure 13: Distribution shift of $\mathbb{E}[\hat{\rho}_a]$ between training and sampling for the LTG backbone model trained with heterogeneous time sampling on the QM9 dataset. Curves are the empirical expectation of the ratio $\hat{\rho}_a$ for each atom type in each chemical formula as a function of time t . **Left:** $\hat{\rho}_a$ computed from the geometric states x_t occurring in the forward process (training). **Right:** $\hat{\rho}_a$ computed from the geometric states x_t occurring during sampling. Only test molecules whose chemical formula has been seen during training are included in the comparison.

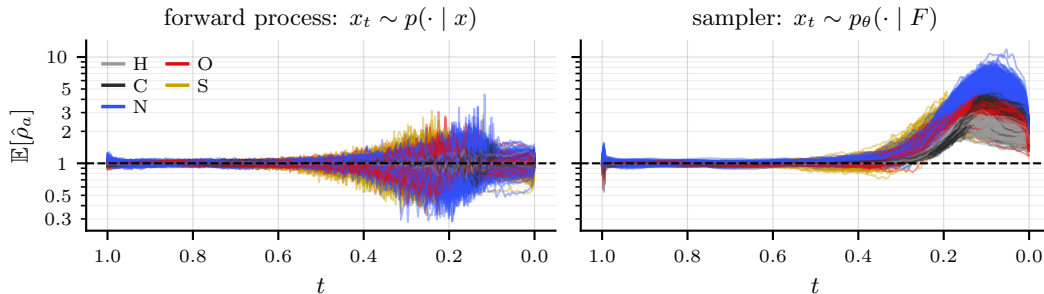


Figure 14: Distribution shift of $\mathbb{E}[\hat{\rho}_a]$ between training and sampling for the standard backbone model on the GEOM-Drugs dataset. Curves are the empirical expectation of the ratio $\hat{\rho}_a$ for each atom type in each chemical formula as a function of time t . **Left:** $\hat{\rho}_a$ computed from the geometric states x_t occurring in the forward process (training). **Right:** $\hat{\rho}_a$ computed from the geometric states x_t occurring during sampling. Only test molecules whose chemical formula has been seen during training are included in the comparison.

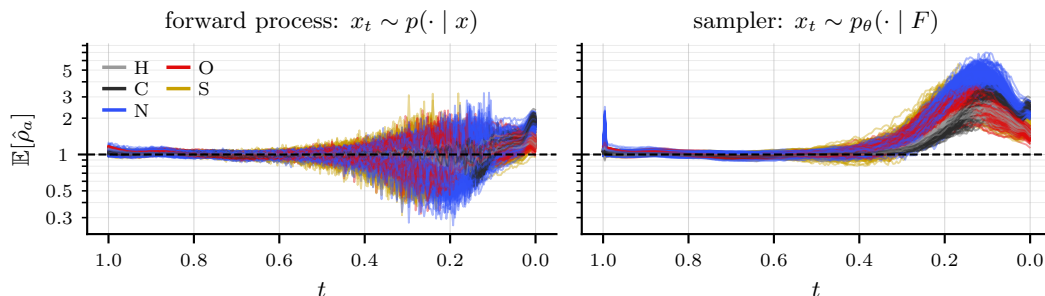


Figure 15: Distribution shift of $\mathbb{E}[\hat{\rho}_a]$ between training and sampling for the LTG backbone model trained with heterogeneous time sampling on the GEOM-Drugs dataset. Curves are the empirical expectation of the ratio $\hat{\rho}_a$ for each atom type in each chemical formula as a function of time t . **Left:** $\hat{\rho}_a$ computed from the geometric states x_t occurring in the forward process (training). **Right:** $\hat{\rho}_a$ computed from the geometric states x_t occurring during sampling. Only test molecules whose chemical formula has been seen during training are included in the comparison.

D.8 Hyperparameter sensitivity analysis

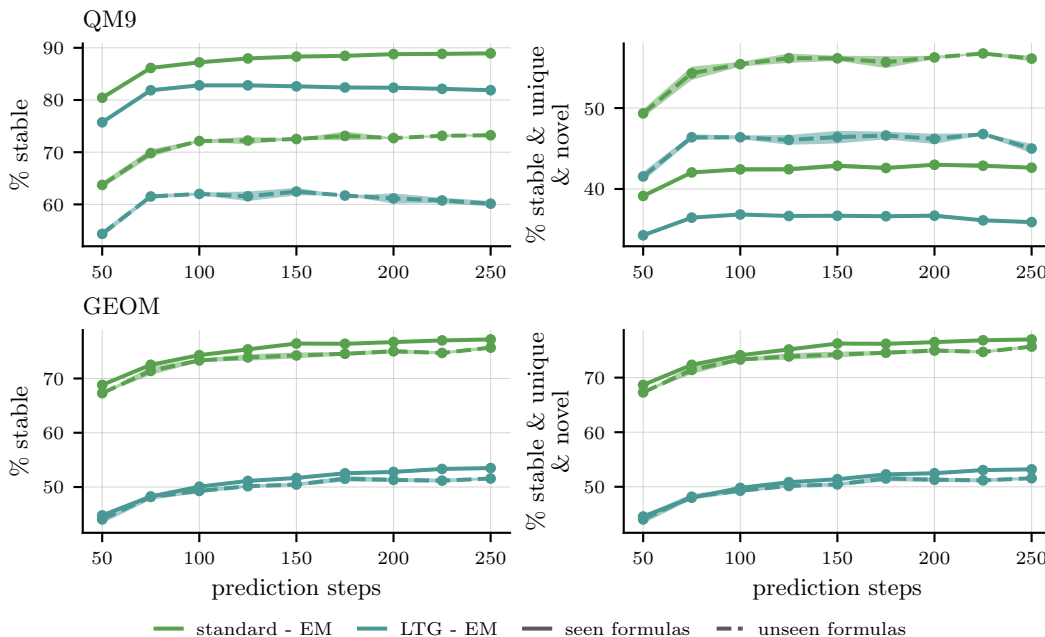


Figure 16: % stable and % stable & unique & novel molecules generated from the validation chemical formulas as the number of integration (prediction) steps is increased in the EM sampler for the two different backbone models (standard and LTG), on QM9 (top) and GEOM-Drugs (bottom). Results are reported as mean ± 1 standard error over three seeds.

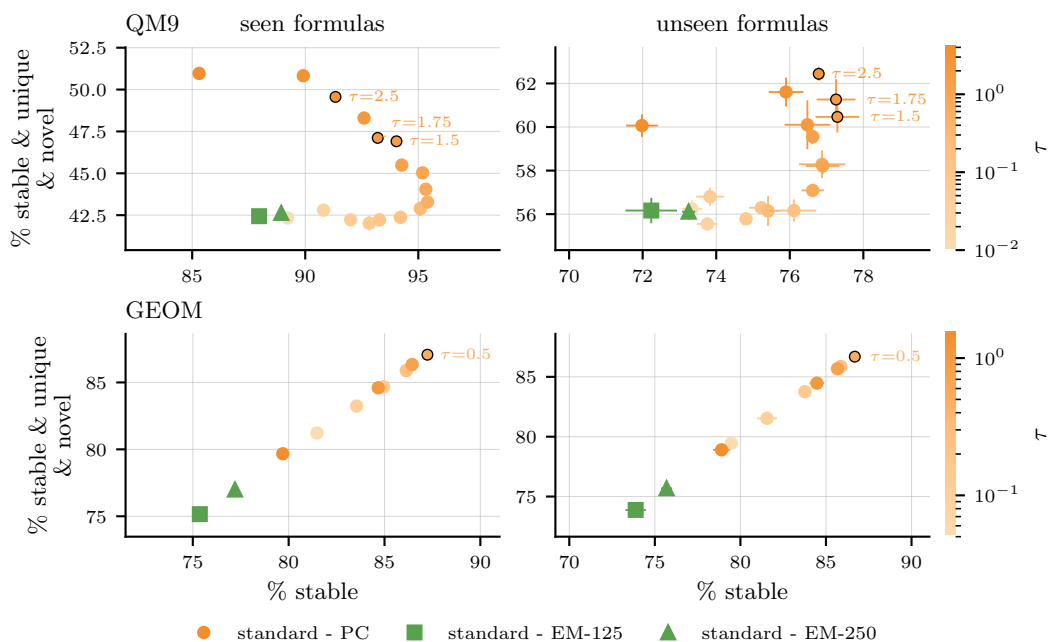


Figure 17: % stable and % stable & unique & novel molecules generated from the validation chemical formulas as the correction step size τ in the PC sampler is varied, on QM9 (top) and GEOM-Drugs (bottom). The two EM baselines are shown for reference. Points are mean ± 1 standard error over three seeds.

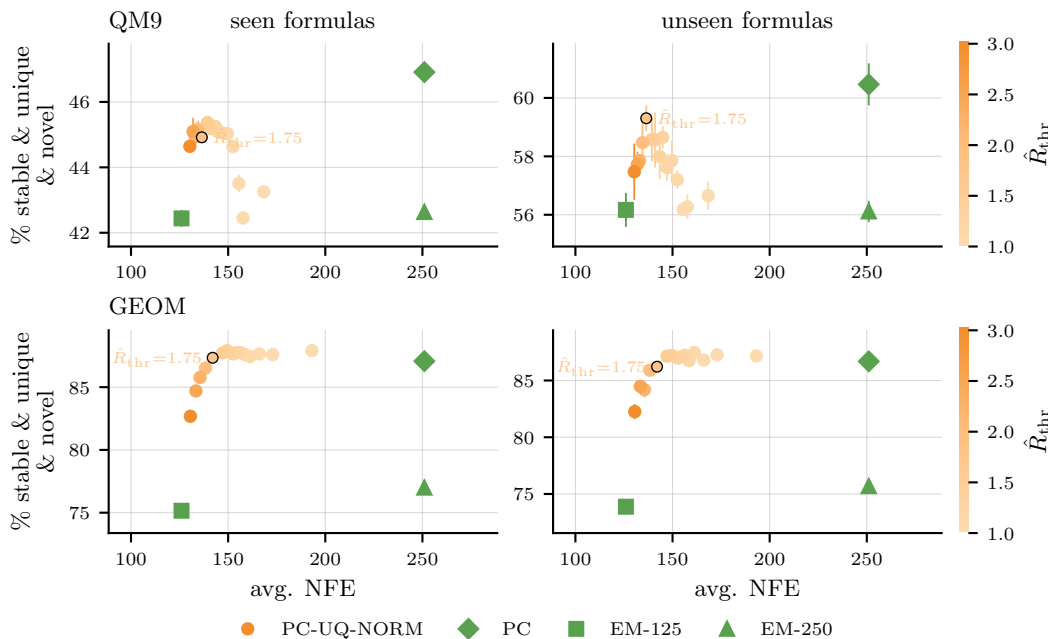


Figure 18: % stable & unique & novel molecules and average NFE per generated molecule for the validation chemical formulas as the correction threshold $\hat{R}_{\text{thr}}$ is varied in the PC-UQ-NORM sampler, on QM9 (top) and GEOM-Drugs (bottom). EM at 125 and 250 steps and the always-on corrector (PC) are shown for reference. Points are mean ± 1 standard error over three seeds.

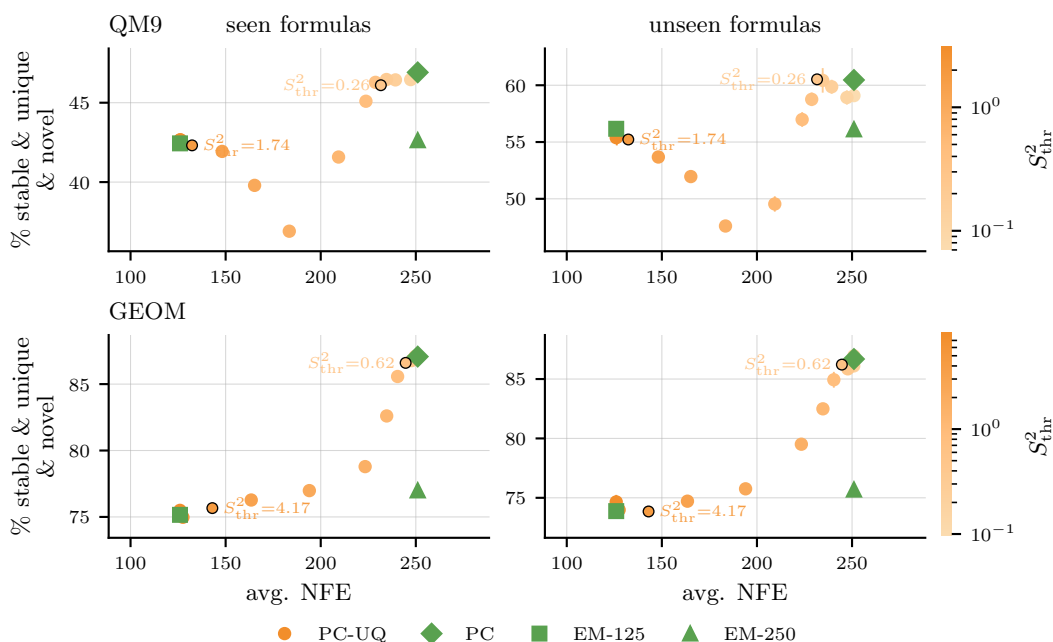


Figure 19: % stable & unique & novel molecules and average NFE per generated molecule for the validation chemical formulas as the correction threshold S^2_{thr} is varied in the PC-UQ sampler, on QM9 (top) and GEOM-Drugs (bottom). EM at 125 and 250 steps and the always-on corrector (PC) are shown for reference. Points are mean ± 1 standard error over three seeds.

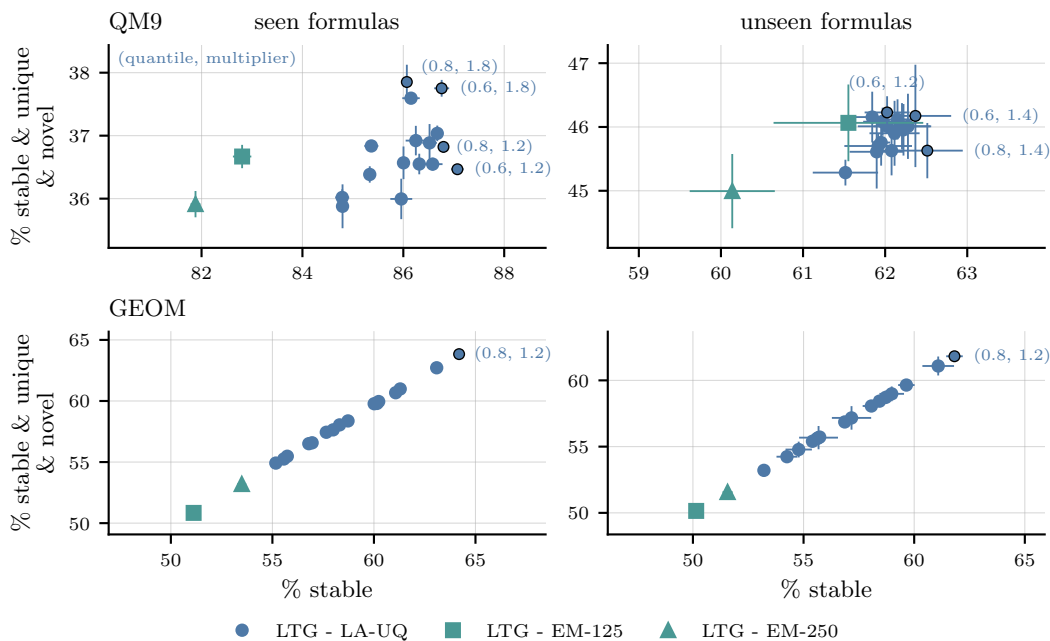


Figure 20: % stable and % stable & unique & novel molecules generated from the validation chemical formulas as the quantile threshold q and look-ahead multiplier κ are varied in the LA-UQ-NORM sampler on the LTG backbone, on QM9 (top) and GEOM-Drugs (bottom). EM at 125 and 250 steps on the same checkpoint are shown for reference. Points are mean ± 1 standard error over three seeds.

Table 7: Settings for every sampler reported in Tabs. 1, 2, 5 and 6. The hyperparameters B , τ , S_{thr}^2 , $\hat{R}_{\text{thr}}$, q , and κ were chosen based on Figs. 16 to 20. Two settings were instead matched rather than swept: the random corrector’s triggering fraction matches the uncertainty trigger’s measured triggering fraction on the same split, and PC-UQ-M’s threshold matches PC-UQ-NORM’s measured cost, which is what makes both NFE-matched controls. Filtering’s keep fraction is the only setting not selected on validation: 0.5 keeps the best half of the generated molecules.

	setting	QM9	GEOM
EM- $\{125, 250\}$	prediction steps B	$\{125, 250\}$	$\{125, 250\}$
PC	prediction steps B	125	125
	corrector step size τ	1.5	0.5
	corrector steps per firing	1	1
PC-RAND-M	prediction steps B	125	125
	corrector step size τ	1.5	0.5
	corrector steps per firing	1	1
	triggering fraction (matched)	0.0837	0.1282
PC-UQ	prediction steps B	125	125
	corrector step size τ	1.5	0.5
	corrector steps per firing	1	1
	trigger threshold S_{thr}^2	0.264	0.620
PC-UQ-M	prediction steps B	125	125
	corrector step size τ	1.5	0.5
	corrector steps per firing	1	1
	trigger threshold S_{thr}^2	1.745	4.172
PC-UQ-NORM	prediction steps B	125	125
	corrector step size τ	1.5	0.5
	corrector steps per firing	1	1
	trigger threshold $\hat{R}_{\text{thr}}$	1.75	1.75
LA-RAND	prediction steps B	125	125
	quantile threshold q	0.6	0.8
	look-ahead multiplier κ	1.4	1.2
LA-UQ	prediction steps B	125	125
	quantile threshold q	0.6	0.8
	look-ahead multiplier κ	1.4	1.2
LA-UQ-NORM	prediction steps B	125	125
	quantile threshold q	0.6	0.8
	look-ahead multiplier κ	1.4	1.2
EM-FILTER(-NORM)	candidates kept	0.5	0.5